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Quiz #1

- ① (f) $\rightarrow$ The lattice type of the crystal, I believe, would be: Trigonal (R) $\rightarrow \alpha = \beta = \gamma < 120^\circ, \neq 90^\circ$

* Remark: This could totally be wrong, and I'm basing this on the fact that we are given two angles in the problem: one @ 60° and the other @ 120° ; $\alpha = \beta = \gamma < 120^\circ$ and @ the same time $\neq 90^\circ$ seemed like a reasonable argument.

② Diamond structure created in VESTA $\rightarrow$ File in email

③ PdO_4 structure created in VESTA $\rightarrow$ File in email

④ CaF_2 structure created in VESTA

$\Rightarrow$ Symmetry Relations for $Fm-3m$:

* Cubic

* Remark: I think that's what is being asked...

⑤ Unit cells for Po (sc), Fe (bcc), and Cu (fcc) created in VESTA

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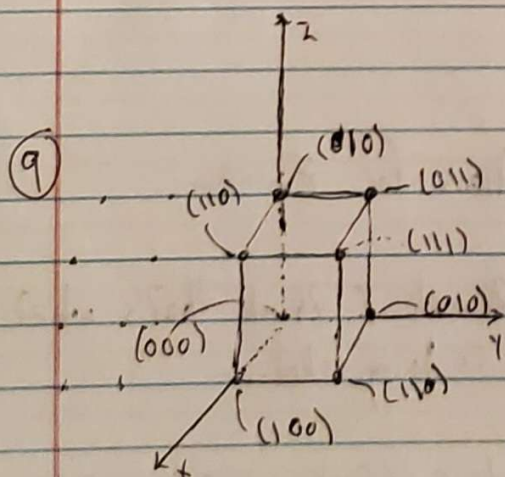
⑥ No Face-Centered Tetragonal Lattice, why?

* Tetragonal crystal system has symmetry of a single four-fold rotation axis

* In a tetragonal lattice with a face-centered unit cell $\rightarrow$ resulting ~~cell~~ cell = $\frac{1}{2}$ the volume of the original lattice $\rightarrow$ This would just be a ~~centered~~ body-centered tetragonal cell. So, face-centered cannot exist.

⑦ $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ bcc does not lie on (112) Plane, $[112]$ crystal direction is perpendicular. $\rightarrow$ Prove it. Honestly, this may be going right over my head, but I am stumped on how to start the proof...

⑧ Family of Planes = planes that are related by symmetry operations of the lattice or the crystal.



The six Miller Indices for all six sides of cubic crystal

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⑩ Distance between (112) plane of a cubic crystal

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$d_{112} = \frac{5.4}{\sqrt{1^2 + 1^2 + 2^2}} = \frac{5.4}{\sqrt{6}} = \underline{\underline{2.20 \text{ \AA}}}$$

⑪ Width of Diffraction Maximum

(a) Show that $|F|^2 = F^* F = \frac{\sin^2 \frac{1}{2} M(a \cdot \Delta k)}{\sin^2 \frac{1}{2} (a \cdot \Delta k)}$

$$* F = \frac{1 - \exp(-iM(a \cdot \Delta k))}{1 - \exp(-i(a \cdot \Delta k))}$$

$$* |F|^2 = \frac{F^* F}{\cancel{F F^*}} = \left[\frac{1 - \exp(-iM(a \cdot \Delta k))}{1 - \exp(-i(a \cdot \Delta k))} \right] * \left[\frac{1 - \exp(iM(a \cdot \Delta k))}{1 - \exp(i(a \cdot \Delta k))} \right]$$

$$* |F|^2 = \frac{(1 - \exp(iM(a \cdot \Delta k)) - \exp(-iM(a \cdot \Delta k)) + \exp(-iM(a \cdot \Delta k)) \exp(i(a \cdot \Delta k)))}{(1 - \exp(i(a \cdot \Delta k)) - \exp(-i(a \cdot \Delta k)) + \exp(-i(a \cdot \Delta k)) \exp(i(a \cdot \Delta k)))}$$

$$* |F|^2 = \frac{2 - ((\cos M(a \cdot \Delta k)) + i \sin M(a \cdot \Delta k) + (\cos M(a \cdot \Delta k) - i \sin M(a \cdot \Delta k)))}{2 - ((\cos(a \cdot \Delta k)) + i \sin(a \cdot \Delta k) + (\cos(a \cdot \Delta k) - i \sin(a \cdot \Delta k)))}$$

$$* |F|^2 = \frac{2 - 2 \cos M(a \cdot \Delta k)}{2 - 2 \cos(a \cdot \Delta k)} = \frac{1 - \cos M(a \cdot \Delta k)}{1 - \cos(a \cdot \Delta k)}$$

$$\boxed{|F|^2 = \frac{\sin^2 \frac{1}{2} M(a \cdot \Delta k)}{\sin^2 \frac{1}{2} (a \cdot \Delta k)}}$$

(b) Show that $\epsilon = 2\pi/M$; ϵ gives position of first zero

* Given that $a \cdot \Delta k = 2\pi h$

* With $\epsilon \rightarrow a \cdot \Delta k = 2\pi h + \epsilon$

* Plugging into $\sin^2 \frac{1}{2} M(a \cdot \Delta k) \rightarrow \sin^2 \frac{1}{2} M(2\pi h/M) = \sin^2 \pi = 0$

* $(a \cdot \Delta k)$ has to equal ϵ , which has to be $2\pi/M$ to equal zero